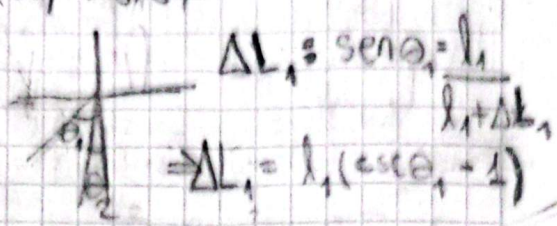


Problemas Agosto 28

1. a) Grados de libertad: $2(x, \dot{x}, y, \dot{y})$



$$\Delta l_1 = \text{sen } \theta_1 = \frac{y}{l_1 + \Delta l_1}$$

$$\Rightarrow \Delta l_1 = l_1 (\cos \theta_1 - 1)$$

Se mueve por todo x, y : $L = T - V = \frac{1}{2} m (\dot{x}^2 + \dot{y}^2) - \frac{1}{2} (k_1 \Delta l_1^2 + k_2 \Delta l_2^2)$

$$\Delta l_1 = \sqrt{x^2 + (y - l_2)^2} - l_1$$

$$\Delta l_2 = \sqrt{y^2 + (x - l_1)^2} - l_2$$

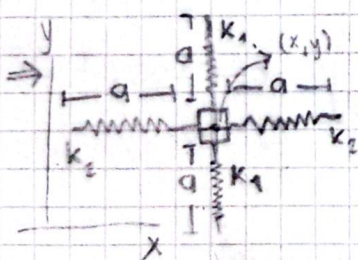
$$\frac{d}{dt} \frac{\partial}{\partial \dot{x}} \left(\frac{1}{2} m \dot{x}^2 \right) - \frac{\partial}{\partial x} \left[-\frac{1}{2} \left(k_1 (\sqrt{x^2 + (y - l_2)^2} - l_1)^2 + k_2 (\sqrt{y^2 + (x - l_1)^2} - l_2)^2 \right) \right]$$

$$= m \ddot{x} + \frac{1}{2} \left[2 k_1 \sqrt{x^2 + (y - l_2)^2} - l_1 \right] \left(\frac{1}{\sqrt{x^2 + (y - l_2)^2}} \right) (x) + k_2 \left[\sqrt{y^2 + (x - l_1)^2} - l_2 \right] \left(\frac{1}{\sqrt{y^2 + (x - l_1)^2}} \right) (x - l_1) = 0$$

$$= m \ddot{x} + k_1 (\sqrt{\alpha} - l_1) (\alpha^{-1/2}) (x) + k_2 (\sqrt{\beta} - l_2) (\beta^{-1/2}) (x - l_1) = 0$$

$$\rightarrow \frac{d}{dt} \left(\frac{\partial}{\partial \dot{y}} \left(\frac{1}{2} m \dot{y}^2 \right) \right) - \frac{\partial}{\partial y} (L) = 0$$

$$= m \ddot{y} + k_1 (\sqrt{\alpha} - l_1) (\alpha^{-1/2}) (y - l_2) + k_2 (\sqrt{\beta} - l_2) (\beta^{-1/2}) (y)$$



$$\sqrt{x^2 + (2a - y)^2} = a + \Delta l_1$$

$$\sqrt{y^2 + (x - a)^2} = a + \Delta l_1'$$

$$\sqrt{x^2 + (y - a)^2} = a + \Delta l_2$$

$$\sqrt{(2a - x)^2 + (y - a)^2} = a + \Delta l_2'$$

$$\Rightarrow V = \frac{1}{2} [k_1 (\Delta l_1 + \Delta l_1') + k_2 (\Delta l_2 + \Delta l_2')]$$